BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

INSTRUMENT LIST AND I/O SCHEDULE

DOCUMENT NO. : BFWH-V1-IN-INL-001

DISCIPLINE : Instrumentation & control

PROJECT : BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

STATUS : Issued for teaching (IFT)

Instrument index and I/O schedule of the E-101 BFW steam heater: 25 lines from BFWH-V1-PID-001 rev D and the DCS of the simulator, 25 I/O (34 installed with 20 % spare), the SIS independence rule, ranging of the FE-101 orifice and the loop typicals. The cause-and-effect chart is BFWH-V1-IN-CE-001 and is not repeated.

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Numbers are taken from the BFW Steam Heater Ver 1 simulator model; values marked assumed / TBC / HOLD are not design data.

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1 Purpose and deliverables

The instrument list turns every bubble on the P&ID into a device with a range, settings, a system (DCS, SIS or local), an I/O type and a fail position. It is the input to the datasheets, the loop diagrams, the I/O list and the cabinet design.

Deliverable	What it holds	For E-101
Instrument index (this document)	Every tag with service, type, range, settings, system, I/O, fail position	Section 2: 25 lines
Instrument datasheets	Process data per instrument: fluid, range, design P/T, materials, connections	Ranges here; process data from the Process engineering tab and the Design tab
Loop diagrams	Wiring from field device to the DCS / SIS card, junction boxes, power	Loop typicals in section 5
I/O list	Count of AI / AO / DI / DO per system, spare and cabinets	Section 3
Cause and effect chart	Which input trips which output (SIS I-100)	BFWH-V1-IN-CE-001 on the C&E matrix tab (not copied here)
Alarm and trip schedule	Alarm and trip settings, delays, priorities	Columns of the list; trips as on the interlock schedule of BFWH-V1-PID-001

2 Instrument list

Tag	Function / loop	Service	Type	Range	Normal	Alarm L / H	Trip	Sys	I/O	Fail	P&ID	Status
FE-101		BFW flow element on 6"-BW-101	Orifice plate, flange taps (β 0.6 assumed)	0–160 m³/h (ΔP see section 4)	—	—	—	Local	—	—	6"-BW-101	existing
FT-101	FIC-101 / FSLL-101	BFW flow to E-101	DP flow transmitter, square root in DCS	0–160 m³/h	100.0	80 / 120	< 30 (5 s, armed in RUN)	DCS	AI	—	6"-BW-101	existing · shared with SIS (note 7)
FT-101S	FSLL-101 (own)	Own low-flow transmitter for FSLL-101 on FE-101 (tag TBC)	DP flow transmitter, own taps	0–120 m³/h	100.0	—	< 30 (5 s)	SIS	AI	—	note 7	recommended · HOLD
FV-101A	FIC-101 OUT	BFW flow control valve	Globe control valve, positioner	0–100 % open	55 % open	—	—	DCS	AO	FO	6"-BW-101	existing
FV-101B	100 − FIC-101 OUT	Minimum-flow recycle to TK-101	Globe control valve (split from FIC-101)	0–100 % open	45 % open	—	—	DCS	AO	TBC	3"-BW-102, note 6	existing · fail position TBC
TT-103	TI-103	BFW inlet temperature	RTD + temperature transmitter	0–40 °C	18.0	12 / 25	—	DCS	AI	—	6"-BW-101	existing
PT-106	PI-106	P-101A/B discharge pressure (auto-start input, HAZOP R01)	Pressure transmitter	0–1,100 kPa g	662	—	—	DCS	AI	—	pump discharge	existing
PDT-109	PDI-109	Tube-side (BFW) ΔP across E-101	DP transmitter	0–120 kPa	60.0	— / 80	—	DCS	AI	—	E-101 tube nozzles	existing
TT-101	TIC-101	BFW outlet temperature (cascade master)	RTD + temperature transmitter	0–120 °C	70.0	67 / 74	—	DCS	AI	—	6"-BW-103	existing
TT-108	TIAH-108 / TSHH-108	Independent BFW outlet temperature	RTD + temperature transmitter	30–100 °C	70.2	— / 85	> 95 (5 s)	SIS	AI	—	6"-BW-103	existing
PT-100	PI-100 / PSHH-100	E-101 shell pressure	Pressure transmitter	−10–260 kPa g	105.0	— / 200	> 250 (3 s)	DCS	AI	—	E-101 shell	existing · shared with SIS (note 7)
PT-100S	PSHH-100 (own)	Own shell-pressure transmitter for PSHH-100 (tag TBC)	Pressure transmitter, compound range (assumed)	−100–400 kPa g	105.0	—	> 250 (3 s)	SIS	AI	—	note 7	recommended · HOLD
TT-100	TI-100	Steam temperature at the shell inlet	RTD + temperature transmitter	90–170 °C	121.3	—	—	DCS	AI	—	14"-LS-102	existing
PT-105	PI-105	Flash steam header pressure upstream of XV-100	Pressure transmitter	0–320 kPa g	195.0	150 / 240	—	DCS	AI	—	12"-LS-101	existing
FE-104		Steam flow element	Orifice / flow nozzle (TBC)	0–16 000 kg/h	—	—	—	Local	—	—	14"-LS-102	existing
FT-104	FIC-104	Flash steam to the shell (cascade slave)	DP flow transmitter, P/T compensation TBC	0–16,000 kg/h	9,935	— / 13,000	—	DCS	AI	—	14"-LS-102	existing
FV-104	FIC-104 OUT	Steam control valve	Globe control valve, positioner	0–100 % open	50 % open	—	—	DCS	AO	FC	14"-LS-102	existing
ZT-104		FV-104 position feedback, deviation alarm vs OUT (HAZOP R10)	Position transmitter	0–100 %	50.0	—	—	DCS	AI	—	FV-104	recommended
XV-100	I-100 output (XY-100)	SIS steam shut-off valve solenoid (de-energise to trip)	On/off valve + solenoid	open / closed	open	—	—	SIS	DO	FC	12"-LS-101, note 4	existing
HS-100	HS-ESD	Emergency shutdown push-button	Push-button (P&ID: on the DCS; hardwired console button recommended)	—	—	—	—	SIS	DI	—	interlock schedule	existing
TT-102	TI-102 / TAL-102	Vent-line temperature at AAV-101 (rev C)	RTD + temperature transmitter	20–140 °C	121.3	L: 10 °C below TI-100	—	DCS	AI	—	1"-VT-101	rev C option

Tag	Function / loop	Service	Type	Range	Normal	Alarm L / H	Trip	Sys	I/O	Fail	P&ID	Status
LT-100		E-101 hotwell level, high alarm below the tube bottom (HAZOP R12)	DP level transmitter	0–25 % of shell ID	5.0	— / 12	—	DCS	AI	—	note 11	recommended · HOLD
LT-101		TK-101 level, low / high alarms and low-low pump trip (HAZOP R02)	Level transmitter	TBC	—	—	—	DCS	AI	—	TK-101	recommended
P-101A/B run		Pump run status from the MCC (assumed interface)	Contact ×2	run / stop	—	—	—	DCS	DI	—	P-101A/B	existing
P-101A/B start		Pump start / stop to the MCC (auto-start, HAZOP R01; assumed)	Contact ×2	—	—	—	—	DCS	DO	—	P-101A/B	existing

Ranges: controller ranges of the model loops and indicator scales of the DCS; normal values: design snapshot; alarms: model alarm table; trips: SIS I-100. Recommended additions (FT-101S, PT-100S, ZT-104, LT-100, LT-101) from HAZOP R02, R10, R12 and P&ID note 7, ranges assumed. Pump status and commands: MCC interface assumed.

3 I/O schedule and SIS independence

System	I/O	Existing	Rev C option	Recommended	Total	Installed (20 % spare)
DCS	AI	9	1	3	13	16
DCS	AO	3	0	0	3	4
DCS	DI	2	0	0	2	3
DCS	DO	2	0	0	2	3
SIS	AI	1	0	2	3	4
SIS	DI	1	0	0	1	2
SIS	DO	1	0	0	1	2
All		19	1	5	25	34

Independence of the SIS inputs

SIS function	Sensor	Also used by	Independent ?	Action
FSLL-101 low-low BFW flow	FT-101	FIC-101 (BPCS control loop)	✗ shared	Own transmitter FT-101S on FE-101, recommended, HOLD (P&ID note 7)
TSHH-108 high-high BFW outlet	TT-108	—	✓ own sensor	Own element, independent of TT-101. TIAH-108 alarm reads the same TT-108: the LOPA tab does not credit both.
PSHH-100 high-high shell pressure	PT-100	PI-100 (BPCS indication and alarm)	✗ shared	Own transmitter PT-100S, recommended, HOLD (P&ID note 7)

Rule: a safety instrumented function reads its own sensor, not the transmitter of the control loop or alarm whose failure it protects against.

4 Ranging the FE-101 orifice and span check

Orifice relation $Q = C\,E\,A_o\,\sqrt{(2\Delta P/\rho)}$, $E = 1/\sqrt{(1-\beta^4)}$. With D = 154.1 mm (6 in sch 40), β = 0.6 (assumed), C = 0.61 (assumed), ρ = 998 kg/m³: d = 92.5 mm, E = 1.0719, ΔP at 160 m³/h = **51.1 kPa** → FT-101 calibrated 0–51.1 kPa, square root in the DCS. Permanent loss at 100 m³/h ≈ $(1-\beta^{1.9})$ ΔP = 12.4 kPa. With a usable flow turndown of 4:1 (assumed) the loop reads from 40 m³/h.

Point	Flow m³/h	% of flow span	ΔP kPa	% of ΔP span	Readable?
Normal	100	62.5	19.98	39.1	✓
Alarm L	80	50.0	12.79	25.0	✓
Alarm H	120	75.0	28.77	56.2	✓
Start permissive	50	31.2	4.99	9.8	✓
Trip FSLL-101	30	18.8	1.80	3.5	✗ (EQ-42)

Position of normal, alarms and trip in the calibrated span (good practice 10–90 %)

Instrument	Range	Normal %	Alarm L %	Alarm H %	Trip %
FT-101	0–160 m³/h	62.5 ✓	50.0 ✓	75.0 ✓	18.8 ✓
FT-101S	0–120 m³/h	83.3 ✓	—	—	25.0 ✓
TT-103	0–40 °C	45.0 ✓	30.0 ✓	62.5 ✓	—
PT-106	0–1,100 kPa g	60.2 ✓	—	—	—
PDT-109	0–120 kPa	50.0 ✓	—	66.7 ✓	—
TT-101	0–120 °C	58.3 ✓	55.8 ✓	61.7 ✓	—
TT-108	30–100 °C	57.4 ✓	—	78.6 ✓	92.9 ✗
PT-100	−10–260 kPa g	42.6 ✓	—	77.8 ✓	96.3 ✗
PT-100S	−100–400 kPa g	41.0 ✓	—	—	70.0 ✓
TT-100	90–170 °C	39.2 ✓	—	—	—
PT-105	0–320 kPa g	60.9 ✓	46.9 ✓	75.0 ✓	—
FT-104	0–16,000 kg/h	62.1 ✓	—	81.2 ✓	—
ZT-104	0–100 %	50.0 ✓	—	—	—
TT-102	20–140 °C	84.4 ✓	—	—	—
LT-100	0–25 % of shell ID	20.0 ✓	—	48.0 ✓	—

5 Loop typicals

- FIC-101: FT-101 on FE-101 → FIC-101 (reverse) → FV-101A (FO) with OUT and FV-101B recycle with 100 − OUT (split, fail position TBC).
- TIC-101 → FIC-104 cascade: TT-101 → TIC-101 (master) sets the remote set point of FIC-104; FT-104 → FIC-104 (slave) → FV-104 (FC).
- SIS I-100: FSLL-101 (via FT-101), TSHH-108 (TT-108), PSHH-100 (via PT-100) and HS-100 → I-100 → XY-100 de-energises → XV-100 closes (FC); FIC-104 to MAN 0 %, TIC-101 to MAN. Pumps keep running.
- Loop design, tuning and modes: Control loop design tab. Trip logic and response: SRS of SIF tab. Cause and effect: BFWH-V1-IN-CE-001.

6 Checks against the model and other documents

Check	This document	Model / other document	Agrees
Every DCS bubble (BUB) tag is in the list	14 of 14 listed	FSLL-101, FIC-101, TI-103, PI-106, PDI-109, TIC-101, TIAH-108, TSHH-108, PI-100, PSHH-100, TI-100, FIC-104, PI-105, TI-102	✓
FT-101 range covers normal, alarms and trip	0–160 m³/h	values 100.0, 80.0, 120.0, 30.0	✓
TT-103 range covers normal, alarms and trip	0–40 °C	values 18.0, 12.0, 25.0	✓
PT-106 range covers normal, alarms and trip	0–1100 kPa g	values 662.4	✓
PDT-109 range covers normal, alarms and trip	0–120 kPa	values 60.0, 80.0	✓
TT-101 range covers normal, alarms and trip	0–120 °C	values 70.0, 67.0, 74.0	✓
TT-108 range covers normal, alarms and trip	30–100 °C	values 70.2, 85.0, 95.0	✓
PT-100 range covers normal, alarms and trip	-10–260 kPa g	values 105.0, 200.0, 250.0	✓
TT-100 range covers normal, alarms and trip	90–170 °C	values 121.3	✓
PT-105 range covers normal, alarms and trip	0–320 kPa g	values 195.0, 150.0, 240.0	✓
FT-104 range covers normal, alarms and trip	0–16000 kg/h	values 9935.4, 13000.0	✓
TT-102 range covers normal, alarms and trip	20–140 °C	values 121.3	✓
Ranges equal the DCS controller / indicator ranges (model)	11 ranges	makeCtrls() and IND agree	✓
FSLL-101 trip setting	30 m³/h	K.tripFlow = 30	✓
TSHH-108 trip setting	95 °C	K.tripT = 95	✓
PSHH-100 trip setting	250 kPa g	K.tripP = 250	✓
FIC-101 alarm settings	L 80 · H 120	ALM L 80 · H 120	✓
TIC-101 alarm settings	L 67 · H 74	ALM L 67 · H 74	✓
TIAH-108 alarm settings	L — · H 85	ALM L — · H 85	✓
PI-100 alarm settings	L — · H 200	ALM L — · H 200	✓
PI-105 alarm settings	L 150 · H 240	ALM L 150 · H 240	✓
FIC-104 alarm settings	L — · H 13000	ALM L — · H 13000	✓
TI-103 alarm settings	L 12 · H 25	ALM L 12 · H 25	✓
PDI-109 alarm settings	L — · H 80	ALM L — · H 80	✓
FV-101A fail position	FO	Control loop design: "fail open"	✓
FV-104 fail position	FC	Control loop design: "fail closed"	✓
XV-100 fail position	FC	P&ID note 4: fails closed	✓
FV-101B fail position defined	TBC	P&ID note 6: "TBC"; not in the model	✗ EQ-40 raised
FSLL-101 low-low BFW flow on its own transmitter (independent of the BPCS)	FT-101	also used by FIC-101 (BPCS control loop)	✗ EQ-41 raised
TSHH-108 high-high BFW outlet on its own transmitter (independent of the BPCS)	TT-108	not used by a BPCS loop	✓
PSHH-100 high-high shell pressure on its own transmitter (independent of the BPCS)	PT-100	also used by PI-100 (BPCS indication and alarm)	✗ EQ-41 raised
FSLL-101 trip readable on FT-101 (orifice turndown)	30 m³/h = 18.8 % flow, 3.5 % ΔP	readable from 40 m³/h (4:1 assumed)	✗ EQ-42 raised
Start permissive FIC-101 ≥ 50 readable on FT-101	31.3 % flow, 9.8 % ΔP	readable from 40 m³/h	✓
Trip settings inside 10–90 % of the transmitter span	FSLL-101 18.8 % · TSHH-108 92.9 % · PSHH-100 96.3 %	outside: TT-108, PT-100	✗ EQ-43 raised
PSHH-100 response within the process safety time	5.75 s (SRS tab)	PST 5 s	✗ EQ-44 raised

7 Engineering queries

EQ	Subject	Query	Status
EQ-40	FV-101B fail position	The minimum-flow recycle valve FV-101B has no fail position on BFWH-V1-PID-001 rev D (note 6: TBC) and none in the model. Fail open protects the pump on loss of air but bypasses BFW from E-101 while FV-101A also fails open; confirm the fail position and add it to the C&E (HAZOP R22).	Open
EQ-41	SIS inputs share BPCS transmitters	FSLL-101 reads FT-101 (the FIC-101 control transmitter) and PSHH-100 reads PT-100 (PI-100). One failed transmitter can cause the upset and blind the trip. Own transmitters FT-101S and PT-100S (tags TBC) wired to the SIS are recommended; HOLD until the SRS / LOPA confirm the SIL credit (P&ID note 7).	Open
EQ-42	FT-101 turndown at the FSLL-101 trip	FT-101 is ranged 0-160 m3/h; the FSLL-101 trip at 30 m3/h is 18.8 % of flow and only 3.5 % of the DP span, below the readable limit of 40 m3/h for an orifice loop (4:1, assumed). Range the own SIS transmitter FT-101S for the trip (e.g. 0-120 m3/h, trip at 25 % flow) or use a meter with a wider turndown.	Open
EQ-43	Trip settings near the top of the span	TSHH-108 at 95 C is 92.9 % of the TT-108 span (30-100 C) and PSHH-100 at 250 kPa g is 96.3 % of the PT-100 span (-10 to 260 kPa g); the PSV-100 set (350, assumed) is off scale. Re-range: TT-108 0-150 C, shell pressure -100 to 400 kPa g (compound, also shows vacuum on cooldown).	Open
EQ-44	PSHH-100 response time	The SRS tab gives a PSHH-100 response of 5.75 s (3 s delay + transmitter and logic + XV-100 stroke 2 s, assumed) against a process safety time of 5 s. Shorten the trip delay, confirm the XV-100 stroke time on its datasheet, and confirm the transmitter response.	Open
EQ-45	XV-100 stroke time and hotwell level	The XV-100 closing time of 2 s is assumed in the model and the SRS; obtain it from the valve datasheet. The E-101 hotwell level is not measured (P&ID note 11): LT-100 with a high alarm below the tube bottom (12 % of shell ID) is recommended (HAZOP R12), HOLD.	Open

Notes

- Numbers come from tools/mk_data.js: unit.json (design snapshot, alarm table, trips) and the live page (DCS ranges IND, makeCtrls), with the same functions as the Instrumentation & control tab.
- Assumed: β 0.6 and C 0.61 for FE-101, 4:1 orifice turndown, 20 % spare, ranges of the recommended transmitters, MCC interface for the pumps. TBC / HOLD: FV-101B fail position (EQ-40), own SIS transmitters (EQ-41), XV-100 stroke and LT-100 (EQ-45).
- PSHH-100 response 5.75 s and process safety time 5 s are taken from the SRS of SIF tab.